

GENERAL THETA FOUNDATIONS I: POSITIVE REALIZATION AND EXACT STREAMING MEMORY

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ABSTRACT. We study the number of retained states needed to realize a specified finite causal experiment. A criterion for simultaneous attainment of the support-component rank is given by enclosing simplices and common positive shift equations. Unlike a residual-vertex criterion, it allows all internal continuations, including those outside the residual hull. For a binary-query channel on n acquired bits, we determine the exponential memory rate at every fixed signal strength $0 < \eta < 1$. The checkpoint, fixed-order streaming, and adaptive-acquisition minima all have rate $1 - h_2((1 + \eta)/2)$. The upper construction realizes every conditional probability exactly and has a fully charged sequential buffer; its excess logarithmic space is $O_\eta(\sqrt{n \log n})$. At row total-variation tolerance ε , the corresponding rate is obtained by replacing η by $(\eta - 2\varepsilon)_+$. We also construct an exact $(t + 1)$ -state online encoder for $\eta \leq (2n - 1)^{-1}$ and determine the sharp $n + 1$ -state threshold $\eta = 1/n$ whenever a Hadamard matrix of order $n + 1$ exists. The entropy exponent is classical in random-access coding; the conclusions here concern exact conditional channel synthesis and the complete retained-state cost. The relation to residual probabilistic automata is made explicit by a finite language with exponentially many residual states and a polynomial-size positive realization.

1. INTRODUCTION

A finite statistical memory is a collection of possible continuations, not a collection of numerical coordinates. A low-dimensional space of predictions can require many positive generators. Conversely, insisting that every internal state be an observed residual can exclude much smaller stochastic realizations. There is a third constraint: positive factorizations at successive cuts must arise from the same causal machine. The distinctions between dimension, positivity, and compatibility are the subject of this paper.

We work with complete finite response arrays, known primitive probabilities, and an explicit list of charged cuts. Arbitrary stochastic rows are atomic operations. Every retained selector, observation buffer, and data-dependent protocol state is included in the register. The program and the declared clock are fixed before observations. These conventions distinguish an exact state-count theorem from a finite-random-bit implementation or an acquired calibration theorem.

1.1. The realization question. At a cut t , normalized residual continuations form a finite set $\mathcal{R}_t$ in a causal polytope $\mathcal{P}_t$. Intersections of their minimal faces partition them into support components. If component j has affine dimension $r_{t,j} - 1$, every stochastic implementation needs at least $s_t = \sum_j r_{t,j}$ states. Residual vertices attain this bound when each component hull is a simplex. That sufficient condition does not describe all rank-saturating realizations.

Theorem 2.3 gives a necessary and sufficient condition for attaining the whole profile s_t . Each component must admit an enclosing simplex with $r_{t,j}$ normalized causal vertices in its affine slice. The vertices need not be residuals. Their one-step restrictions must satisfy common nonnegative

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shift equations with the next cut's vertices. Thus the condition separates static enclosure from simultaneous causal realization. For explicit algebraic arrays it becomes a finite existential system over the real numbers. This is an exact equality criterion, not a polynomial-time algorithm for a succinct controller or a solution of autonomous shared-row minimization.

1.2. The growing experiment. An input $x \in \{-1, 1\}^n$ is acquired one coordinate at a time. After all coordinates have been acquired, a query $j \in [n]$ is supplied. The required binary response is

$$(1) \quad \mathbb{P}(B = b \mid x, j) = \frac{1 + b\eta x_j}{2}, \quad b \in \{-1, 1\}.$$

The signal η is specified, and the law is required for every input and query. Write K_n for the smallest state count at the query barrier, W_n^{ad} for the smallest whole-schedule peak with adaptive acquisition, and W_n^{fo} for the fixed-order peak. In the latter two quantities all persistent choices during acquisition are charged. They satisfy $K_n \leq W_n^{\text{ad}} \leq W_n^{\text{fo}}$.

Let h_2 be binary entropy in bits and put

$$(2) \quad I(u) = 1 - h_2((1 + u)/2), \quad 0 \leq u \leq 1.$$

The main quantitative conclusion, Theorem 4.4, is

$$(3) \quad \lim_{n \rightarrow \infty} \frac{\log_2 K_n(\eta)}{n} = \lim_{n \rightarrow \infty} \frac{\log_2 W_n^{\text{ad}}(\eta)}{n} = \lim_{n \rightarrow \infty} \frac{\log_2 W_n^{\text{fo}}(\eta)}{n} = I(\eta) \quad (0 < \eta < 1).$$

The checkpoint upper error is $O_\eta(\log n)$ bits. The streaming error is $O_\eta(\sqrt{n \log n})$ bits. A fixed codebook is chosen in the existence proof, so there is no persistent public random seed hidden from the state count. For every rowwise total-variation tolerance ε , Theorem 5.1 gives the rate $I((\eta - 2\varepsilon)_+)$.

The exponent in (3) is not new as a random-access coding exponent. Ambainis, Nayak, Ta-Shma and Vazirani [1, 2] prove the matching first-order communication rate for success probability at least $(1 + \eta)/2$. We prove a version fixing *each* conditional response in (1), and then account for the largest retained register during a one-pass implementation. A success inequality by itself is not an exact conditional channel. Likewise, a short message by itself is not a space bound for computing that message from a stream. The proofs below keep both distinctions.

Finite-dimensional information is also obtained. An elementary online simplex construction gives the exact acquisition profile $t + 1$ for $0 < \eta \leq (2n - 1)^{-1}$, improving the smaller signal regime of the preceding development. Any $n + 1$ -state checkpoint factorization requires $\eta \leq 1/n$. If a Hadamard matrix of order $n + 1$ exists, a different online encoder attains that entire interval with peak $n + 1$ (Theorem 3.4). In particular, the threshold is sharp on the infinite sequence $n = 2^k - 1$. The latter construction has peak $n + 1$; it is not claimed to have only $t + 1$ states at every earlier cut.

1.3. One notation and resource convention.

Symbol	Meaning
$C, R_h, \mathcal{P}_t$	Complete causal law, normalized residual, and ambient causal polytope.
$r_{t,j}, s_t$	Rank of support component j , and their sum.
$B_{t,j,k}$	An internal normalized continuation, not necessarily a residual.
$K_n, W_n^{\text{fo}}, W_n^{\text{ad}}$	Query-cut, fixed-order peak, and adaptive-acquisition peak state counts.
η, ε, I	Specified signal, uniform row-TV tolerance, and entropy rate in bits.

State counts are cardinalities; their binary memory cost is the ceiling of their base-two logarithm. The query is an atomic current input and is answered without retaining an uncharged copy. Input-dependent scheduling information has to be retained. A terminal output can use two labels; this constant does not affect any rate. An externally scheduled overwrite may draw fresh independent randomness, but past randomness survives only through the retained label.

1.4. Relationship with the preceding foundations development. The causal-experiment, predictive-quotient, minimax-contact, and physical comparison results of the preceding development [10] are preserved in a separate complete mathematical companion. The present article proves its new claims without asking the reader to reconstruct a proof from that volume. Section 7 identifies the relations with positive realization, residual automata, and random-access coding. The old known-target physical memory theorems remain unchanged; neither a new fully adaptive collision theorem nor closure of the historical analytic pipeline is inferred from (3).

2. COHERENT POSITIVE REALIZATIONS AT THE RANK BOUND

2.1. Finite causal arrays. Fix a finite horizon and finite action and report alphabets. A complete causal array $C(\mathbf{a} \mid \mathbf{y})$ is nonnegative and normalized, and its action-prefix marginal depends only on preceding reports. A terminal output is an action with a dummy subsequent report. All storage cuts required by the interface are included. In particular, an action which must be remembered while awaiting a report has a charged buffer cut; it is not available later as an uncharged input. Legal operation and buffer-copy requirements can be specified as zeros and mandatory readouts in the complete array. Elementary phases may be split to expose them. The results below concern the resulting fixed, clocked interface.

For a controller prefix h of positive conditional mass, let R_h be its normalized array of future actions, with future reports supplied externally. Let $\mathcal{R}_t$ be all distinct such arrays at cut t and $\mathcal{P}_t$ the polytope of normalized causal arrays on its remaining interface. Neither object is a runtime oracle. At zero-mass prefixes no residual is defined; the complete array still specifies which continuations have zero mass. Partial almost-sure specifications require a choice, or an optimization, of complete extensions and are a different input to the theorem.

For $B \in \mathcal{P}_t$, let $p_B(a)$ be its next-action marginal, independent of the unread report. Let $S_{a,y}B$ be the unnormalized restriction obtained by fixing that action and the next report. If $p_B(a) > 0$, $S_{a,y}B/p_B(a) \in \mathcal{P}_{t+1}$; at zero probability the restriction is zero. For an actual residual, each positive normalized restriction is again an actual residual. These maps are linear before normalization.

Write $F_t(R)$ for the minimal face of $\mathcal{P}_t$ containing R . Join two members of $\mathcal{R}_t$ when their minimal faces intersect, and let $\mathcal{R}_{t,j}$, $1 \leq j \leq c_t$, be the connected components. Define

$$(4) \quad r_{t,j} = 1 + \dim \text{aff } \mathcal{R}_{t,j}, \quad s_t = \sum_{j=1}^{c_t} r_{t,j}.$$

Normalization places each component in an affine hyperplane not through zero, so $r_{t,j}$ is also its ordinary linear row rank.

Lemma 2.1 (Separated-generator bound). *Every stochastic realization has at least s_t states at cut t . If it has exactly s_t , the states used by component j are precisely $r_{t,j}$ affinely independent points of $\mathcal{P}_t \cap \text{aff } \mathcal{R}_{t,j}$.*

Proof. Starting the machine from a state s defines a normalized continuation $B_s \in \mathcal{P}_t$. Conditioning on a prefix gives

$$R_h = \sum_s E_h(s) B_s, \quad E_h(s) \geq 0, \quad \sum_s E_h(s) = 1.$$

Every positively used B_s belongs to $F_t(R_h)$, by the face property. Consequently the sets of states used by different support components are disjoint. The convex hull of the states used by component j contains $\mathcal{R}_{t,j}$, and hence has at least $r_{t,j}$ generators. This proves the lower bound. Equality in the sum leaves exactly that many generators in each component and no additional used states. Containment of an affine space of dimension $r_{t,j} - 1$ in their affine span forces independence and equality of the two affine spans. No residual-state restriction has been imposed. $\square$

Definition 2.2 (Coherent enclosing simplices). A coherent rank enclosure consists, at every cut and component, of $r_{t,j}$ points $B_{t,j,k} \in \mathcal{P}_t \cap \text{aff } \mathcal{R}_{t,j}$ whose convex hull contains $\mathcal{R}_{t,j}$, together with nonnegative numbers $w_{t,j,k,a,y,u}$, where u ranges over all vertices selected at the next cut, satisfying

$$(5) \quad S_{a,y} B_{t,j,k} = \sum_u w_{t,j,k,a,y,u} B_{t+1,u},$$

$$(6) \quad \sum_u w_{t,j,k,a,y,u} = p_{B_{t,j,k}}(a).$$

The initial array belongs to the initial selected hull. At the terminal cut there is the single empty continuation. Action-buffer phases obey the same equations on their explicitly split interface.

The enclosure points are automatically affinely independent: they contain a set of the full required affine dimension. In particular, “simplex” describes an enclosing object, not the residual hull.

Theorem 2.3 (Rank saturation and causal compatibility). *For a complete finite clocked causal array, the following are equivalent:*

- (1) *A stochastic realization has profile $K_t = s_t$ at every charged cut.*
- (2) *A coherent rank enclosure exists.*

If every component residual hull is a simplex, its own vertices give such an enclosure. The equivalence itself places no simpliciality condition on the residual hulls.

Proof. Suppose first that a rank-saturating machine exists. Delete unused states. Lemma 2.1 assigns its continuations to the indicated component affine slices and proves their independence and enclosure property. From a given state the actual next-action law is p_B . Conditional on action a and report y , the actual update probabilities $q(u \mid B, a, y)$ are common to all histories

giving that state. Take $w_u = p_B(a)q(u \mid B, a, y)$. The law of the future program gives (5) and (6). Thus the equations hold for one simultaneous collection, including buffer cuts.

Conversely, use the selected vertices as state labels. Initialize by an enclosing convex decomposition of C . From state B , draw the next action according to p_B . When $p_B(a) > 0$, update after report y using $w_u/p_B(a)$. At zero $p_B(a)$ the row is unused and can be assigned arbitrarily. Equation (6) makes each update stochastic, while p_B is independent of the unread report. Backward induction in (5) shows that the future array from state B is exactly B . The initialized law is therefore C . The number of labels is s_t at every cut.

For the final assertion, choose the vertices of each residual hull. They are actual residuals because the residual set is finite. Their positive normalized restrictions are residuals at the next cut and have convex decompositions in that cut's chosen vertices. Multiplying these decompositions by the action marginal supplies the required weights. This is the residual construction, now a special case of the equivalence. $\square$

Corollary 2.4 (Algebraic recognition). *For a fully listed rational or real-algebraic array, simultaneous attainment of s_t is decidable by a finite existential system of polynomial equalities and inequalities. A positive answer admits real-algebraic stochastic rows. Here algebraic input numbers are given by integer defining polynomials and isolating intervals.*

Proof. The finite residuals, component graphs, affine slices, and ranks are computed by exact linear algebra and linear feasibility. Write the unknown enclosure vertices in these affine slices and impose the linear causal-polytope constraints. Introduce convex coefficients for every residual enclosure, and weights for (5)–(6). All unknown products have degree at most two. Independence is forced by enclosure of the known full-dimensional component, so no extra determinant condition is needed. Real quantifier elimination, for example cylindrical algebraic decomposition [3, 4], decides this finite system. A nonempty semialgebraic set defined over the real algebraic numbers has a real algebraic point, which the same sample-point procedure supplies. The construction in the theorem then gives algebraic rows. $\square$

This is not a complexity bound polynomial in the horizon or in a succinct program. Nor does it replace equalities identifying a stochastic row at different autonomous times. Those equalities are extra constraints. The theorem characterizes rank saturation, not arbitrary minimum positive rank. Its essential additional condition over a list of static enclosures is the common shift equation.

2.2. What static rank does and does not measure. Let A be a nonnegative row-stochastic matrix with one output alphabet. Let $K(A)$ be the least number of normalized output rows whose convex hull contains its rows. Let $K_{\text{aff}}(A)$ impose the additional condition that generators lie in the affine hull of the rows, and let $K_{\text{res}}(A)$ require generators to be actual rows.

Proposition 2.5 (The normalization boundary). *For this single-output model,*

$$(7) \quad \text{rank } A \leq \text{rank}_+ A = K(A) \leq K_{\text{aff}}(A) \leq K_{\text{res}}(A) = |\text{ext conv}\{A_i\}|.$$

Writing $r = \text{rank } A$, equality $K(A) = r$ holds if and only if an $(r - 1)$ -simplex in the probability simplex contains every row and lies in their affine hull. For a family of queried output laws, the same convex characterization uses the product of the output simplices; ordinary nonnegative rank of the concatenated matrix is only a lower bound unless its factors satisfy all block-normalization equations.

Proof. A stochastic encoder-decoder factorization is a nonnegative factorization. Conversely, if $A = UV$ is nonnegative, discard zero rows of V and put $d_s = \sum_y V_{sy} > 0$. Then $B_{sy} = V_{sy}/d_s$ is stochastic and $E_{is} = U_{is}d_s$ has row sum one because A does. This proves $\text{rank}_+ A = K(A)$. The

remaining inequalities follow by restricting the generators. The extreme points of a finite hull are precisely the necessary actual-row generators, proving the last equality. If r generators enclose affine dimension $r - 1$, their affine hull must be the row affine hull and they are independent. This proves the simplex criterion and its converse.

With a query j , a physical state has a separate normalized decoder row for every j . Normalizing the total of a concatenated generator only fixes the sum over all blocks; it does not make every block sum one. The admissible ambient set is therefore the product of simplices, which proves the stated distinction. $\square$

The single-output affine-slice restriction is the transpose-normalized form of restricted non-negative rank [7]. At one cut, positive generator complexity also decomposes additively over the support components: a generator cannot serve two components by the proof of Lemma 2.1, and independent component decompositions can be united. For one output alphabet these components are the disjoint support blocks of the matrix. This static additivity is not asserted to solve the compatibility problem between different cuts.

Proposition 2.6 (A nonsimplicial equality and a strict inequality). *For the four binary-query mean rows $(\pm a, \pm a)$, ordinary normalized row rank is three. If $0 < a \leq 1/3$, their positive generator number is three although their residual hull is a square. If $1/2 < a < 1$, their positive generator number is four although all conditional probabilities are strictly positive.*

Proof. The rank assertion follows from affine dimension two. The triangle with vertices $(-a, -a), (3a, -a), (-a, 3a)$ lies in the outer square when $a \leq 1/3$ and contains the inner square; its defining inequalities are $z_i \geq -a$ and $z_1 + z_2 \leq 2a$. This proves the first assertion. For the second, put $d_x(z) = \frac{1}{2} \sum_{i=1}^2 (1 - x_i z_i)$ for $x \in \{-1, 1\}^2$. A convex representation of ax uses a generator with $d_x(z) < 1/2$, because its average is $1 - a < 1/2$. Such strict caps are disjoint for distinct x : a differing coordinate contributes one to $d_x + d_{x'}$. Four distinct generators are required, and the four rows themselves suffice. This proves a full-support rank gap without assuming residual-state realizations. $\square$

3. EXACT FINITE STREAMING AT THE RANK THRESHOLD

3.1. Checkpoint and streaming models. For $n \geq 1$ and $0 \leq \eta \leq 1$, the channel to be synthesized is (1). Every input coordinate is acquired exactly once before the query. In a fixed-order encoder the order is $1, \dots, n$. In an adaptive encoder the next unread coordinate can depend on the retained state and fresh randomness; any information required to make that choice legal must be retained. There is no readable transcript or unrecorded schedule. After acquisition, all input-dependent information passes through a single finite state before the external query arrives. One query is answered; a joint law for simultaneous answers to all queries is not required.

The checkpoint minimum $K_n(\eta)$ imposes no bound on upstream memory. The quantities $W_n^{\text{fo}}(\eta)$ and $W_n^{\text{ad}}(\eta)$ impose a bound on every cut, including intermediate compression and retained random selectors. The clock in the fixed-order convention is prescribed externally. A terminal binary label counts as at most two states. The query is a current atomic input; a model requiring its storage has to charge that storage separately.

Lemma 3.1 (Cube enclosure). *For $\eta > 0$, $K_n(\eta)$ is the smallest number of points $z_s \in [-1, 1]^n$ whose convex hull contains $[-\eta, \eta]^n$. In particular $K_n(\eta) \geq n + 1$. This lower bound and every lower bound obtained from such an enclosure also apply to adaptive acquisition.*

Proof. From state s , let $z_{s,j}$ be the mean binary output under query j . Conditioning on the complete input gives an encoder distribution E_x and hence

$$(8) \quad \eta x = \sum_s E_x(s) z_s.$$

Conversely, such a representation followed by a binary decoder with mean $z_{s,j}$ realizes the prescribed channel. Upstream memory is unrestricted for this converse construction. Convexity reduces enclosure of the inner cube to its corners. Its affine dimension gives $n + 1$. For adaptive acquisition, conditioning on the complete input still gives one distribution of the retained state; the subsequent decoder is common to all inputs. This uses the acquisition barrier, not an assertion about arbitrary interactive scheduling. $\square$

3.2. An online simplex with a linear signal window.

Theorem 3.2 (An exact minimal acquisition profile). *For every $n \geq 1$ and $0 < \eta \leq (2n - 1)^{-1}$,*

$$(9) \quad K_n(\eta) = W_n^{\text{ad}}(\eta) = W_n^{\text{fo}}(\eta) = n + 1.$$

One fixed-order encoder has exactly $t + 1$ states after t acquired bits, $0 \leq t \leq n$, and this profile is minimal at each acquisition cut for that order. Its stochastic acquisition rows are rational and independent of η .

Proof. For $t \geq 1$ use labels $0, 1, \dots, t$ with associated vectors

$$(10) \quad v_{t,0} = -\eta \mathbf{1}, \quad v_{t,i} = -\eta \mathbf{1} + 2t\eta e_i \quad (1 \leq i \leq t).$$

Their convex hull is

$$P_t = \{z \in \mathbb{R}^t : z_i \geq -\eta, \sum_i z_i \leq t\eta\}.$$

Every vertex lies in the outer cube because $(2t - 1)\eta \leq 1$. Moreover $P_t \times \{-\eta, \eta\} \subset P_{t+1}$. The barycentric coordinates of $w \in P_t$ are

$$\lambda_i(w) = \frac{w_i + \eta}{2t\eta}, \quad \lambda_0(w) = \frac{t\eta - \sum_i w_i}{2t\eta}.$$

They are nonnegative and sum to one.

Here is an explicit update avoiding any stored real vector. After t inputs, the state distribution is

$$(11) \quad \mathbb{P}(S_t = i \mid x_{1:t}) = \frac{\mathbf{1}_{\{x_i=1\}}}{t}, \quad \mathbb{P}(S_t = 0 \mid x_{1:t}) = \frac{|\{i \leq t : x_i = -1\}|}{t}.$$

To acquire bit x_{t+1} , keep the old label with probability $t/(t + 1)$. With probability $1/(t + 1)$ replace it by $t + 1$ when $x_{t+1} = 1$ and by zero when $x_{t+1} = -1$. For $t = 0$ only the replacement is used. Induction proves (11). The mean of v_{t,S_t} is exactly $\eta x_{1:t}$. At the query barrier use the binary decoder with mean $v_{n,S_n,j}$. This proves exact synthesis. Only the label survives an update. The last raw bit and the replacement coin do not persist in an auxiliary register.

The final lower bound follows from Lemma 3.1. At an earlier fixed-order cut, fix the unread bits identically for all prefixes. The future response vectors for queries among the first t coordinates contain the t -dimensional cube of means $\eta x_{1:t}$. Factoring these continuations through the current state requires at least $t + 1$ labels. Thus the entire acquisition profile is minimal. The final binary output uses no more labels than the peak $n + 1$. $\square$

For $t \geq 2$ the actual prefix residuals have a cube, not a simplex, as their hull. Equations (10) and (11) are therefore a growing nonsimplicial family of coherent rank enclosures. Theorem 2.3 applies without requiring that an internal vertex be a residual.

3.3. The sharp $n + 1$ -state boundary in Hadamard dimensions.

Lemma 3.3 (A simplex obstruction). *If $K_n(\eta) = n + 1$ with $\eta > 0$, then $\eta \leq 1/n$.*

Proof. The enclosing vertices $v_0, \dots, v_n \in [-1, 1]^n$ are affinely independent. Write their barycentric affine functions as $\lambda_s(z) = a_s + b_s \cdot z$. Nonnegativity on the inner cube gives $a_s \geq \eta \|b_s\|_1$, and $\sum_s a_s = 1$. The identity $z = \sum_s \lambda_s(z) v_s$ gives $\sum_s v_s b_s^T = I_n$. Taking its trace yields

$$n = \sum_s b_s \cdot v_s \leq \sum_s \|b_s\|_1 \leq \eta^{-1}.$$

This proves the assertion for arbitrary enclosing vertices, not just a chosen simplex construction. $\square$

Theorem 3.4 (A sharp threshold with an online realization). *Suppose that a Hadamard matrix of order $m = n + 1$ exists. For $\eta > 0$,*

$$(12) \quad K_n(\eta) = W_n^{\text{ad}}(\eta) = W_n^{\text{fo}}(\eta) = n + 1 \iff \eta \leq 1/n.$$

This applies in particular to every $n = 2^k - 1$, $k \geq 1$. More generally, a Hadamard matrix of order $m \geq n + 1$ gives an m -state streaming realization whenever $0 < \eta \leq 1/n$.

Proof. Normalize the first column of the Hadamard matrix to one, and select n other columns. Let $v_s \in \{-1, 1\}^n$, $1 \leq s \leq m$, be their rows. Orthogonality gives

$$(13) \quad \frac{1}{m} \sum_s v_{s,j} = 0, \quad \frac{1}{m} \sum_s v_{s,i} v_{s,j} = \mathbf{1}_{\{i=j\}}.$$

For each index i and observed bit x_i , define the probability law

$$q_{i,x_i}(s) = \frac{1 + x_i v_{s,i}}{m}.$$

Its mean vector is $x_i e_i$ by (13). At step i , retain the preceding state with probability $(i - 1)/i$; otherwise redraw a label from q_{i,x_i} . The first step always draws. After t bits its state law is $t^{-1} \sum_{i \leq t} q_{i,x_i}$. Consequently $\mathbb{E}[v_{S_n} | x] = x/n$. The final decoder with mean $n\eta v_{S_n,j}$ is legal for $\eta \leq 1/n$ and gives mean ηx_j . This is an online machine with m labels; the chosen index of a redraw is not separately retained. All acquisition probabilities are rational.

For $m = n + 1$, Lemma 3.1 proves optimality and Lemma 3.3 excludes larger signals. The Sylvester construction $H_1 = (1)$ and $H_{2m} = \begin{pmatrix} H_m & H_m \\ H_m & -H_m \end{pmatrix}$ gives the claimed infinite sequence directly. $\square$

The equality in (12) concerns the peak, not every intermediate width. The encoder in this proof may have $n + 1$ possible labels already after its first bit. In contrast, Theorem 3.2 attains the individual $t + 1$ bounds in its smaller signal interval. Choosing the next power of two for m also gives a less than $2(n + 1)$ -state upper bound at $\eta \leq 1/n$ for arbitrary n ; it does not decide the exact simplex threshold in every dimension.

4. EXACT SYNTHESIS THROUGHOUT THE FIXED-SIGNAL RANGE

All logarithms in rates are base two. Natural logarithms are denoted by $\ln$. Define I by (2); thus $I(0) = 0$, $I(1) = 1$, and I is increasing on $[0, 1]$.

4.1. A uniform corner-cap lower bound.

Lemma 4.1 (Entropy cap). *For $z \in [-1, 1]^n$ and $0 \leq u \leq 1$, the fraction of sign vectors x with $x \cdot z \geq nu$ is at most $2^{-nI(u)}$. Therefore*

$$(14) \quad K_n(\eta) \geq 2^{nI(\eta)}.$$

Proof. For a uniform sign vector X and $t \geq 0$,

$$\mathbb{E} e^{tX \cdot z} = \prod_{i=1}^n \cosh(tz_i) \leq \cosh(t)^n.$$

Exponential Markov inequality gives the bound $\exp\{-n \sup_{t \geq 0} (tu - \ln \cosh t)\}$. For $0 < u < 1$ the maximum occurs at $t = \operatorname{arctanh} u$. Substitution gives $\sup (tu - \ln \cosh t) = (\ln 2)I(u)$. The two endpoints follow by limits. If a set of generators encloses ηx , then some generator z_s has $x \cdot z_s \geq n\eta$. These caps therefore cover all 2^n sign vectors. The fraction bound and the union bound imply (14). $\square$

The exponent in this lemma is the classical random-access lower exponent [1, Theorem 2.1]. Its cap proof is useful here because it applies directly to exact and perturbed conditional means.

4.2. An enclosing codebook with exact, not one-sided, responses. For $0 < \eta < 1$ and $0 < \delta \leq (1 - \eta)/2$, put

$$(15) \quad k = \left\lceil \frac{n(1 + \eta + 2\delta)}{2} \right\rceil, \quad r = \frac{2k}{n} - 1,$$

$$(16) \quad M(n, \eta, \delta) = \left\lceil \frac{n+1}{\delta} \{n \ln(1 + 2/\delta) + 1\} 2^{nI(r)} \right\rceil.$$

Here $k \leq n$ and $\eta + 2\delta \leq r \leq \eta + 2\delta + 2/n$.

Lemma 4.2 (A Hamming-layer enclosure). *There is a set of at most $M(n, \eta, \delta)$ sign vertices whose convex hull contains $[-\eta, \eta]^n$. In particular,*

$$(17) \quad K_n(\eta) \leq \min\{2^n, M(n, \eta, \delta)\}.$$

For every fixed $0 < \eta < 1$ there is a constant C_η such that

$$(18) \quad nI(\eta) \leq \log_2 K_n(\eta) \leq nI(\eta) + C_\eta \log_2(n+1) \quad (n \geq 1).$$

Proof. Fix $a \in \mathbb{R}^n$ with $\|a\|_1 = 1$. Orient each coordinate by the sign of a_i , choosing either orientation at zeros. Condition a uniform sign vertex V on having exactly k coordinates aligned with these orientations. Under this conditional law $Z = a \cdot V \leq 1$ and $\mathbb{E}Z = r$. If $p = \mathbb{P}(Z \geq \eta + \delta \mid k)$, then

$$r \leq p + (1 - p)(\eta + \delta), \quad p \geq \frac{r - \eta - \delta}{1 - \eta - \delta} \geq \delta.$$

It follows that, for an unconditional uniform sign vertex,

$$(19) \quad \mathbb{P}(a \cdot V \geq \eta + \delta) \geq \delta \binom{n}{k} 2^{-n} \geq \frac{\delta}{n+1} 2^{-nI(r)} =: q.$$

For the second inequality, k is a mode of a binomial distribution with success probability k/n , so its probability is at least $1/(n+1)$. Rewriting that probability gives $\binom{n}{k} \geq 2^{nh_2(k/n)}/(n+1)$. The endpoints $k = 0, n$ have the same interpretation by continuity.

Choose a δ -net of the ℓ^1 unit sphere of size at most $(1 + 2/\delta)^n$. Such a net follows by taking a maximal separated set and comparing the volumes of disjoint ℓ^1 balls of radius $\delta/2$ with the ball of radius $1 + \delta/2$. Draw $M = M(n, \eta, \delta)$ independent uniform sign vertices. For any net

direction the probability that none has scalar product at least $\eta + \delta$ is at most e^{-Mq} . The probability that some net direction fails is at most $(1 + 2/\delta)^n e^{-Mq} \leq e^{-1} < 1$. Hence a fixed choice of vertices succeeds on every net direction. Their support function is one-Lipschitz in the ℓ^1 norm, so it is at least η on the entire unit sphere. Homogeneity and convex separation imply that their hull contains the inner cube. Repeated vertices can be discarded. Full retention gives the other upper bound.

For fixed η , take $\delta = 1/n$ once $n \geq 2/(1 - \eta)$. Then $r \leq \eta + 4/n$, and the derivative $I'(u) = \frac{1}{2} \log_2((1+u)/(1-u))$ is bounded on a fixed neighborhood of η lying below one. Thus $nI(r) \leq nI(\eta) + O_\eta(1)$ for sufficiently large n . The remaining factor in (16) is polynomial up to a logarithm and contributes $O(\log(n+1))$ bits. Enlarging C_η handles the finitely many smaller n . Combine this with Lemma 4.1. $\square$

The hull containment gives a convex representation of *each* ηx , and hence exactly the means in (1). It does not merely ensure that a coordinate is correct with probability at least $(1 + \eta)/2$. Once a successful vertex set has been chosen, all its probabilities are fixed in the program. No random choice of a codebook survives at runtime. For rational η , the enclosure linear systems admit rational encoder coefficients; for algebraic η they admit coefficients in the corresponding ordered field. The proof is an existence and finite-synthesis result, not a claim of fast codebook construction.

4.3. From a terminal codebook to a one-pass machine.

Lemma 4.3 (Charged block realization). *Suppose the ℓ -coordinate channel at signal η has a checkpoint codebook with M generators. Then, for $1 \leq \ell \leq n$,*

$$(20) \quad W_n^{\text{fo}}(\eta) \leq \max\{2, 2^\ell M^{\lfloor n/\ell \rfloor}\}.$$

All conditional output probabilities are realized exactly.

Proof. Split the fixed input order into $b = \lfloor n/\ell \rfloor$ full blocks and a final remainder of length less than ℓ . While acquiring a block, retain the previously chosen block labels and its current raw prefix. Upon completing a block, draw one label using that block's exact encoder, overwrite its raw buffer, and retain only the new label with the previous labels. The encoder table is a fixed row of the declared program. A remainder is retained raw.

At a cut the state is a tuple of at most b codebook labels and at most ℓ raw bits. During the overwrite the old buffer is replaced atomically, not retained alongside a new private selector. Thus the number of possible labels is bounded as in (20). The compression epochs are prescribed clock phases; they do not depend on observations.

For a query in a complete block, use the corresponding label's decoder coordinate. Its conditional mean is ηx_j independently of the other labels. For a query in the remainder, draw a binary response of mean ηx_j from the retained bit. At most two terminal labels are required. The label sampling uses fresh randomness, and all its persistent effect is in the tuple. No unread bit, chosen codebook, or previous random seed is a free register. $\square$

Theorem 4.4 (Exact fixed-signal streaming rate). *For every fixed $0 < \eta < 1$, there exists $C'_\eta < \infty$ such that*

$$(21) \quad nI(\eta) \leq \log_2 K_n(\eta) \leq \log_2 W_n^{\text{ad}}(\eta) \leq \log_2 W_n^{\text{fo}}(\eta),$$

$$(22) \quad \log_2 W_n^{\text{fo}}(\eta) \leq nI(\eta) + C'_\eta \sqrt{n \log_2(n+1)}.$$

Consequently all three normalized logarithmic complexities converge to $I(\eta)$, as in (3). The checkpoint has the stronger error bound (18).

Proof. The lower bounds have already been proved and survive adaptive acquisition. For sufficiently large n , choose $\ell = \lceil \sqrt{n \log_2(n+1)} \rceil \leq n$. Lemma 4.2 supplies a block codebook with $\log_2 M \leq \ell I(\eta) + C_\eta \log_2(\ell + 1)$. Lemma 4.3 gives, after increasing an absolute constant,

$$\log_2 W_n^{\text{fo}} \leq \ell + \frac{n}{\ell} \{ \ell I(\eta) + C_\eta \log_2(\ell + 1) \} \leq n I(\eta) + O_\eta(\sqrt{n \log(n+1)}).$$

Again the finitely many smaller n are absorbed into C'_η . Division by n proves the limits. $\square$

This theorem covers every fixed interior signal, rather than just two endpoint windows. It does not determine every finite integer value of K_n or W_n , their lower-order difference, or a uniform sharp law when η varies with n . Equations (16) and (20) remain explicit nonasymptotic bounds in those varying-signal problems. In the strong finite regime $\eta > 1 - 1/n$, the disjoint-cap argument gives $K_n = W_n^{\text{ad}} = W_n^{\text{fo}} = 2^n$: the encoder average of $\frac{1}{2} \sum_i (1 - x_i z_i)$ is below $1/2$, and its strict caps for distinct input corners are disjoint. Full retention matches this bound. That sharper endpoint statement is consistent with, but not needed for, Theorem 4.4.

5. UNIFORMLY APPROXIMATE EXPERIMENTS AND THEIR EXACT RATE

For two binary-output channels on the same input/query set, define

$$d_{\text{row}}(P, \tilde{P}) = \max_{x,j} \text{TV}(P(\cdot | x, j), \tilde{P}(\cdot | x, j)).$$

For $\varepsilon \geq 0$, this is a uniform comparison of experiments, not an error averaged over a selected input prior. Let $K_n^\varepsilon(\eta)$, $W_n^{\text{ad},\varepsilon}(\eta)$ and $W_n^{\text{fo},\varepsilon}(\eta)$ denote the corresponding minima allowing $d_{\text{row}} \leq \varepsilon$. Put $\kappa = (\eta - 2\varepsilon)_+$.

Theorem 5.1 (A sharp row-error memory rate). *If $0 < \kappa < 1$, then*

$$(23) \quad 2^{nI(\kappa)} \leq K_n^\varepsilon(\eta) \leq K_n(\kappa), \quad W_n^{\text{fo},\varepsilon}(\eta) \leq W_n^{\text{fo}}(\kappa).$$

In particular, for fixed $\eta \in (0, 1]$ and fixed $0 \leq \varepsilon < \eta/2$ with $\kappa < 1$, all three normalized logarithmic approximate complexities converge to $I(\kappa)$. For $\varepsilon \geq \eta/2$, one query state and at most two terminal states suffice, so the rate is zero. At $(\eta, \varepsilon) = (1, 0)$ the exact state counts equal 2^n .

Proof. Let $m_j(x)$ be the mean output of an admissible approximate channel. Binary total variation is half the absolute difference of means; hence $|m_j(x) - \eta x_j| \leq 2\varepsilon$. Factoring the approximate channel through states gives $m(x) = \sum_s E_x(s) z_s$, with $z_s \in [-1, 1]^n$. Therefore

$$x \cdot m(x) \geq n(\eta - 2\varepsilon) = n\kappa$$

when $\kappa > 0$. Some used generator has $x \cdot z_s \geq n\kappa$. The caps in Lemma 4.1 cover the input corners, giving the lower bound, also for adaptive acquisition.

Conversely, exactly synthesize the channel with mean κx_j . Its distance from the required one is $(\eta - \kappa)/2 = \varepsilon$. This proves the upper bounds and allows Theorem 4.4 to be applied at the fixed signal κ . If $\kappa = 0$, the fair-output channel is independent of input and query and has row defect $\eta/2$. It needs one state before output. Finally, an exact noiseless response to every query determines every input bit through the stored generator; the cap argument at $u = 1$, or disjoint supports, gives 2^n states, matched by retention. $\square$

5.1. A statistical consumer: reusable binary sensors. Consider a parameter $x \in \{-1, 1\}^n$ and a preparation interface that reveals its coordinates during acquisition. A later single coordinate measurement has conditional mean ηx_j . A simulator must be installed before the measurement index is chosen. Theorem 4.4 is an exact state-cost theorem for this simulator, not a theorem about a prior chosen to fit a codebook. Theorem 5.1 gives its uniform error-memory exponent, with the same lower bound against adaptive acquisition.

Corollary 5.2 (Comparison under downstream experiments). *A simulator with row defect ε changes the expectation of any common $[0, 1]$ -valued decision loss, after one queried report and arbitrary parameter-independent postprocessing, by at most ε , uniformly in x and in a randomized query distribution independent of the simulator's hidden state. If a sequence of fresh installations has conditional row defect at most ε at each round and is coupled to the same downstream history-dependent controller, the T -round path-law defect is at most $\min(1, T\varepsilon)$.*

Proof. At one round, mixture over the common query and contraction under postprocessing do not increase total variation; bounded losses then have expectation difference at most ε . For repeated installations, couple the two executions only while their public histories agree. The downstream action laws then agree and the chance of a first discrepant report in each active round is at most ε . A union bound proves the path statement. The conditional hypothesis is essential; it is ensured, for example, by fresh independent reinstallation and fresh sampling randomness at every round. $\square$

A single compressed label need not support a sequence of independent queries with the product output law. Observing an answer may change the conditional distribution of that label. Thus repeated use of the same unrefreshed code is not justified by the one-query theorem. If all n noisy answers are required jointly and independently, the specified product channel has full matrix rank 2^n for every $\eta > 0$, and needs at least 2^n query-barrier states. Indeed its transition matrix is the n -fold Kronecker power of the invertible binary symmetric channel. The distinction between a later chosen measurement and a joint vector of measurements is operational, not a change of notation.

The transport argument in the corollary is classical total-variation contraction and coupling. Its quantitative memory input is the new exact-synthesis law. It is a local statistical consumer, not a claim that a Fourier, large-deviation, or nonlinear-semigroup theorem in a different part of the foundations program has been proved by finite channel synthesis.

6. RESIDUAL AUTOMATA AND ARBITRARY POSITIVE STATES

For a stochastic language L on finite words and a prefix u with $L(u\Sigma^*) > 0$, its normalized residual is $L_u(v) = L(uv)/L(u\Sigma^*)$. A residual probabilistic automaton requires every state's generated language to be one of these residuals. An arbitrary probabilistic automaton need not satisfy that restriction. The distinction is explicit in the probabilistic residual line initiated by Esposito, Lemay, Denis and Dupont [6], and in [5, Section 2 and Propositions 16, 19]. Canonical minimality within a residual representation class is consequently not a minimum over all positive generators.

The following construction makes the distinction within the precise streaming family, without appealing to a conjectured difference between models. It concerns the total number of states of a generative automaton; its clock is explicitly incorporated in the states.

Theorem 6.1 (A finite-language separation). *Fix $n \geq 2$ and rational $0 < \eta \leq (2n - 1)^{-1}$. There is a finite stochastic language $L_{n,\eta}$ such that every residual probabilistic automaton generating it has at least 2^n states, whereas an unrestricted rational probabilistic automaton has at most*

$$(24) \quad \frac{(n+1)(n+2)}{2} + n(n+1) + 1$$

states. The lower bound holds already for the states used immediately before the query symbol.

Proof. Use distinct symbols u_-, u_+ for acquired bits, q_j for queries, and o_-, o_+ for answers. Give mass

$$(25) \quad L_{n,\eta}(u_{x_1} \cdots u_{x_n} q_j o_b) = \frac{2^{-n}}{n} \frac{1 + b\eta x_j}{2}$$

to those words and zero to every other word. The masses sum to one. After a length- n supported prefix, the residual has probabilities $(1 + b\eta x_j)/(2n)$ on $q_j o_b$. These 2^n residuals are an injective affine image of the corners of an n -cube. In particular, every one is an extreme point of their convex hull.

In a probabilistic realization, condition on such a prefix. The residual is a convex combination of the state languages then occupied. Since it is supported on words of length two of the displayed query-answer form, nonnegativity forces every positively used state language to have that same support restriction. In a residual automaton that state language must itself be a residual of $L_{n,\eta}$. A residual after a supported prefix of length t is concentrated on length $n + 2 - t$; hence a residual concentrated on length two must come from $t = n$. There are no other supported prefix types at that length. Therefore the used state languages belong to the 2^n cube-corner residuals above. Extremality forces each corner to occur as a state language. They are distinct, proving the lower bound.

For the upper bound, generate the u -symbols as independent fair bits. Simultaneously run the encoder of Theorem 3.2; a transition labelled u_x and updating to s' has probability $\frac{1}{2}T_{t,x}(s, s')$. Include the phase t in the state. The acquisition phases $t = 0, \dots, n$ have a total of $\sum_{t=0}^n (t+1) = (n+1)(n+2)/2$ states. At phase n , emit q_j uniformly and enter a state retaining (s, j) ; there are at most $n(n+1)$ such states. Emit o_b with the exact decoder probability and enter one common terminating state. Every transition and terminal probability is nonnegative rational, and the generated language is (25). This proves (24) with its clock and query buffer included. $\square$

The theorem does not assert that exponential succinctness differences between probabilistic models are new. Its role is to identify a concrete instance in which a residual-only minimum cannot be substituted for the all-machine minimum in this paper. In the present example the latter has a genuinely online positive realization. The state languages of that realization generally lie outside the residual hull.

One can equivalently treat the query as an external input, recovering the checkpoint and streaming models of Section 3. Turning it instead into a generated symbol in (25) is what permits a direct comparison with stochastic-language automata. The factor $1/n$ is essential: it normalizes the query choice rather than appending an unnormalized family of conditional laws.

7. THEOREM BOUNDARIES AND FURTHER QUESTIONS

7.1. What is inherited from realization theory. The probabilistic residual finite-state automata paper of Esposito–Lemay–Denis–Dupont [6] is an essential predecessor, not a peripheral citation. Its announced characterization uses finite generation by residual stochastic languages and canonical minimal forms. The later full treatment [5, Propositions 16, 19] provides a unique inclusion-minimal residual generating set and the finite-generation characterization by residual probabilistic automata. The relevant restriction is on the state languages. Our explicit comparison in Theorem 6.1 uses that restriction rather than inferring an all-machine minimum from the word “canonical.” The publisher record and abstract of [6], and the full cited sections of [5], are the bases of this comparison; we do not claim a completed line-by-line audit of the 2002 chapter.

Positive stochastic realization, including stable polyhedral sets of continuations, is classical [8, 12]. The stationary HMM existence question in [12] is not the same input as a fully listed finite controlled array. Nevertheless the principle that positive generators must be closed under appropriate shifts is shared. Theorem 2.3 should therefore not be read as an invention of positive realization. It makes rank saturation, affine-slice enclosure, and the simultaneous finite shift constraints explicit. Its use here is the non-residual streaming construction, the sharp

Hadamard-dimensional peak threshold, and the charged exact-synthesis bounds, not an assertion that a finite normal form alone settles the general realization problem.

Gillis–Glineur [7, Definition 1 and Theorem 1] distinguish restricted nonnegative rank from unrestricted nonnegative rank and relate the former to nested polytopes. Proposition 2.5 is the normalized single-output specialization, with the transpose convention made explicit. A queried law has block-normalization constraints, and coherent realization has constraints between cuts; these must not be removed when applying a static matrix theorem.

Compatible coverings for incompletely specified sequential machines [11] concern a different form of missing information. Our array theorem uses complete laws and allows support zeros in those laws; it does not choose a canonical completion of a partial law. Filtered-experiment comparison [9] is an adjacent comparison framework, not a result replaced by the constructions here. No general non-overlap assertion about that framework is made. The precise remaining source-access limits are recorded with the revision response.

7.2. What is inherited from random-access coding. The classical lower and upper communication rates in [1, Theorems 2.1 and 2.2] are $(1 - h_2(p))n$ and $(1 - h_2(p))n + O(\log n)$ for recovery success at least p . Their upper construction uses a finite family of masks and permutations to make the guarantee hold for every input and coordinate. Thus the entropy function and the polynomial-overhead communication scale are classical; neither is claimed as a new discovery here.

Our exact-channel proof uses a support-function enclosure of the entire inner cube. It supplies exact prescribed conditional probabilities, not only a success inequality. The block construction then prices all retained raw data and selected codewords. The limiting exponent remains the classical one, but the operational conclusion is about exact channel synthesis at every row and the peak of the whole streaming schedule. The proof of Theorem 5.1 shows precisely how a uniform row-error tolerance changes that exponent. It does not claim a result for arbitrary sequentially repeated queries on one installation.

7.3. Precision and implementation. Atomic stochastic rows can contain a specified real parameter. A rational row can be sampled exactly by a variable-length fair-bit procedure, but that statement alone does not give a bounded-time or bounded-workspace sampler. The present state counts concern atomic rows. The finite codebooks and rows may be enormous as program objects; program description length and computational time are not included in K_n or W_n . This is the same distinction as in the preceding known-target physical statements. Acquiring an unknown target and approximating its stochastic row requires calibration data, retained counters, and precision-dependent workspace. Those results are retained with their original costs in the mathematical companion [10].

7.4. Conclusions. The proved exact class is now described by the coherent-enclosure criterion, which is necessary and sufficient for the support-component rank profile to be attained. Residual-hull simpliciality is one sufficient subcase. The growing query family has nonsimplicial residual hulls, an explicit minimal acquisition profile in a linear signal window, and a sharp $n + 1$ -state threshold in Hadamard dimensions.

Unrestricted positive-state minimization, autonomous row sharing, and joint optimization over all controlled policies remain different problems. We have not supplied an example in which every static cut attains its rank bound but no simultaneous rank profile exists; the criterion states the exact compatibility test rather than claiming such a separation without proof. The historical geometric and unbounded-operator obligations of the larger program are likewise not consequences of a finite-state coding theorem.

The former fixed-interior-signal gap in the query family is resolved at the level of its exponential rate, both exactly and at a fixed uniform row-error tolerance. Finite-size checkpoint versus streaming gaps, optimal lower-order terms, the sharp simplex threshold in all dimensions, and critical regimes with signal varying with n are not identified by these limits. The explicit enclosure and block bounds provide quantitative statements in those regimes without replacing them by an unsupported exact phase diagram.

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